

CES and Isoelastic Energy Demand Models

Equivalence, Conversion, and Partial Adjustment

Working Paper — Draft

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1. Executive Summary

This paper describes two ways of modelling how fuel consumption responds to price changes—the **CES** (constant elasticity of substitution) form and the **isoelastic** (constant elasticity) form—and shows that they are locally equivalent. It then describes how to introduce **partial adjustment** dynamics, which govern how quickly the economy moves toward the equilibrium implied by new prices.

The key results are:

- The CES is a *structural* model that generates an internally consistent $n \times n$ elasticity matrix from just two parameters (σ, η) plus observed cost shares. The isoelastic form is the *reduced-form* first-order approximation to the CES.

- For small price changes (under $\sim 20\%$ of baseline), the two forms give nearly identical results. For large price changes, the CES is more accurate because it re-linearises as shares evolve.
- Partial adjustment is a layer on top of either form. It introduces an adjustment speed $\lambda \in (0, 1]$ per sector, creating a distinction between short-run and long-run elasticities.
- The adjustment speed can be calibrated directly from the ratio of short-run to long-run elasticities reported in the empirical literature: $\lambda \approx \sigma^{\text{SR}}/\sigma^{\text{LR}}$.
- For the Total Effective Carbon Price (TECP) metric, the long-run equilibrium is the appropriate reference point and partial adjustment is not needed. For year-by-year emissions projections, partial adjustment is essential.

The CES provides the disciplined parameter structure; the isoelastic form provides analytical tractability for the TECP formula; and partial adjustment provides the dynamic realism for forecasting. The three work in concert.

2. Two Ways of Writing the Same Behaviour

2.1 The CES form

The CES is a *structural* model. A sector chooses a bundle of fuels to deliver energy services at minimum cost, subject to a CES aggregator. The solution gives fuel quantities as a function of all prices simultaneously:

$$q_f = q_f^0 \left(\frac{p_f}{p_f^0} \right)^{-\sigma} \left(\frac{C}{C^0} \right)^{\sigma+\eta} \quad (1)$$

where $C = \left(\sum_f \alpha_f p_f^{1-\sigma} \right)^{1/(1-\sigma)}$ is the CES cost index, σ is the substitution elasticity, η is the demand elasticity, and α_f are calibrated share parameters. The key feature is that C couples all fuels together: changing the price of one fuel affects the quantities of every other fuel through the cost index.

2.2 The isoelastic form

The isoelastic model is a *reduced-form* description. It writes the response of each fuel directly in terms of own-price and cross-price elasticities:

$$\frac{\Delta q_f}{q_f} \approx \varepsilon_{ff} \frac{\Delta p_f}{p_f} + \sum_{g \neq f} \varepsilon_{fg} \frac{\Delta p_g}{p_g} \quad (2)$$

where ε_{ff} is the own-price elasticity and ε_{fg} are cross-price elasticities. This is a first-order log-linear approximation: each fuel responds to every price with a constant elasticity, and the total response is additive.

2.3 Local equivalence

The CES *generates* a specific pattern of isoelastic elasticities. Taking the log-differential of the CES quantity equation:

$$\varepsilon_{ff} = -\sigma(1 - s_f) + (\sigma + \eta)s_f = -\sigma + (\sigma + \eta)s_f \quad (3)$$

$$\varepsilon_{fg} = (\sigma + \eta)s_g \quad \text{for } g \neq f \quad (4)$$

where $s_f = p_f q_f / \sum_g p_g q_g$ is the cost share. Equation (3) decomposes the own-price elasticity into a substitution effect ($-\sigma$, always negative) and a cost-index effect $((\sigma + \eta)s_f)$. Equation (4) says all cross-price elasticities are proportional to the other fuel's cost share.

The CES is therefore a *disciplined* isoelastic model: it generates an entire $n \times n$ elasticity matrix from just two parameters (σ and η) plus observed cost shares. An unconstrained isoelastic model would require n^2 parameters. The CES achieves parsimony by imposing:

- **Symmetry**: cross-elasticities are related via cost shares (consistent with Slutsky conditions).
- **Proportionality**: all cross-elasticities from a given fuel are proportional to that fuel's cost share.
- **Adding up**: Engel and Cournot aggregation are satisfied automatically.

2.4 When they diverge

For small price changes (under $\sim 20\%$ of base price), the isoelastic approximation to the CES is very accurate. For large price changes, the two forms diverge because:

- The CES properly tracks the nonlinear reallocation of cost shares as prices change.
- The isoelastic form holds shares fixed at the linearisation point.
- As fuel shares approach zero or one, the CES naturally reflects diminishing substitution possibilities; the isoelastic form does not.

2.5 When to use which

	CES	Isoelastic
Spreadsheet implementation	Needs cost index C	Direct per-fuel formulas
Large price changes	Accurate	Approximation degrades
Cross-elasticity consistency	Guaranteed	Must be imposed manually
Number of parameters	2+ shares	n^2 (or 2+ shares if from CES)
Analytical tractability	Cost function available	Simpler algebra
Partial adjustment	Natural	Natural

For the TECP formula (Paper I), we use the isoelastic (semi-elasticity) form because we are linearising around the baseline. This is exact for small policy perturbations and avoids computing C inside the formula. But the elasticities we plug in come from the CES, so the two forms work in concert: the CES provides discipline, the isoelastic form provides simplicity.

3. Converting Between Forms

3.1 CES to isoelastic elasticities

Given CES parameters σ_s , η_s , and cost shares s_{fs} , the isoelastic elasticity matrix is:

$$\varepsilon_{ff,s} = -\sigma_s + (\sigma_s + \eta_s) s_{fs} \quad (5)$$

$$\varepsilon_{fg,s} = (\sigma_s + \eta_s) s_{gs} \quad (g \neq f) \quad (6)$$

3.2 Isoelastic elasticities to semi-elasticities

Semi-elasticities give the proportional response to an absolute (not proportional) price change:

$$\tilde{\varepsilon} = \frac{\varepsilon}{p} \quad (7)$$

3.3 CES directly to semi-elasticities

Combining the above in one step, and incorporating the emissions-relevant transformation:

Intensive semi-elasticity (fuel f , sector s):

$$\tilde{\varepsilon}_{fs}^{\text{int}} = \sigma_s (1 - w_{fs}) \frac{e_f - \bar{e}_s}{p_{fs}} \quad (8)$$

Extensive semi-elasticity (sector s):

$$\tilde{\varepsilon}_s^{\text{ext}} = |\eta_s| \frac{\bar{e}_s}{C_s} \quad (9)$$

Start from		Arrive at
CES parameters (σ_s, η_s)	$\xrightarrow{\text{\S 3.1}}$	Isoelastic elasticities $(\varepsilon_{ff,s}, \varepsilon_{fg,s})$
Isoelastic elasticities	$\xrightarrow{\text{\S 3.2: } \div p}$	Semi-elasticities $(\tilde{\varepsilon}_{fs}^{\text{int}}, \tilde{\varepsilon}_s^{\text{ext}})$
Semi-elasticities + policy prices	$\xrightarrow{\text{Paper I}}$	Total effective carbon price τ^*

4. Partial Adjustment

Both the CES and isoelastic forms describe *long-run equilibrium* responses—the quantities that would prevail if the economy had fully adjusted to new prices. In practice, adjustment takes time: capital stocks must turn over, infrastructure must be built, and habits must change.

4.1 The partial adjustment model

In any year t , actual quantities move a fraction λ of the way from last year's actual toward this year's equilibrium target:

$$q_{fs,t} = q_{fs,t-1} + \lambda_s (q_{fs,t}^* - q_{fs,t-1}) \quad (10)$$

where $q_{fs,t}^*$ is the equilibrium quantity from either the CES or isoelastic form, and $\lambda_s \in (0, 1]$ is the adjustment speed for sector s . This is equivalent to an exponential smoothing filter with half-life $t_{1/2} \approx \ln 2 / \lambda_s$.

4.2 Equilibrium targets

CES form:

$$q_{fs,t}^{\text{CES}} = q_{fs,t-1} \left(\frac{p_{fs,t}}{p_{fs,t-1}} \right)^{-\sigma_s} \left(\frac{C_{s,t}}{C_{s,t-1}} \right)^{\sigma_s + \eta_s} \quad (11)$$

where $C_{s,t} = \left(\sum_f \alpha_{fs} p_{fs,t}^{1-\sigma_s} \right)^{1/(1-\sigma_s)}$.

Isoelastic form:

$$q_{fs,t}^{\text{iso}} = q_{fs,t-1} \prod_g \left(\frac{p_{gs,t}}{p_{gs,t-1}} \right)^{\varepsilon_{fg,s}} \quad (12)$$

Both compute the target relative to the *previous year's actuals*, not the original baseline. This means each year's starting point reflects the accumulated partial adjustment from all prior years.

4.3 CES vs isoelastic under partial adjustment

The one substantive difference between the forms under partial adjustment is that the CES *re-linearises* at each year's actual position (cost shares update endogenously through C), while the isoelastic form uses fixed elasticities. For year-to-year steps with moderate price changes, the difference is small. Over long horizons with transformational price changes (e.g., near-complete electrification), the CES is substantially more reliable because it captures the diminishing marginal substitution as shares approach corner solutions.

4.4 Split adjustment speeds

The intensive and extensive margins may adjust at different rates. To accommodate this, decompose the equilibrium into a fuel-mix component and a demand component:

$$Q_{s,t}^* = \sum_f q_{fs,t}^* \quad (\text{equilibrium total energy}) \quad (13)$$

$$w_{fs,t}^* = q_{fs,t}^* / Q_{s,t}^* \quad (\text{equilibrium energy shares}) \quad (14)$$

Partially adjust each margin separately:

$$Q_{s,t} = Q_{s,t-1} + \lambda_s^{\text{ext}} (Q_{s,t}^* - Q_{s,t-1}) \quad (15)$$

$$w_{fs,t} = w_{fs,t-1} + \lambda_s^{\text{int}} (w_{fs,t}^* - w_{fs,t-1}) \quad (16)$$

Actual quantities:

$$q_{fs,t} = w_{fs,t} \cdot Q_{s,t} \quad (17)$$

This ensures shares sum to one at every time step and allows fuel switching and demand reduction to proceed at different speeds. Switching between gasoline and diesel is fast (λ^{int} near 1); switching from fossil fuels to electricity requires new vehicles or power plants (λ^{int} perhaps 0.1). Demand reduction in electricity may be faster than in transport (λ^{ext} varies by sector).

5. Calibrating Adjustment Speed from the Literature

After a permanent price change at $t = 0$, the cumulative adjustment by year T is:

$$\frac{q_{f,T} - q_{f,0}}{q_f^* - q_{f,0}} = 1 - (1 - \lambda)^T \quad (18)$$

The *effective elasticity at horizon T* is therefore:

$$\sigma_T^{\text{eff}} = \sigma [1 - (1 - \lambda)^T] \quad (19)$$

Most empirical energy studies report separate short-run (1-year) and long-run elasticities. This provides a direct calibration:

$$\lambda \approx \frac{\sigma^{\text{SR}}}{\sigma^{\text{LR}}} \quad \text{and} \quad \sigma = \sigma^{\text{LR}} \quad (20)$$

The same relationship holds for η and λ^{ext} . For example, if the literature reports a short-run price elasticity of gasoline demand of -0.3 and a long-run elasticity of -0.7 , then $\lambda \approx 0.3/0.7 \approx 0.43$ and $\eta = -0.7$. The implied half-life is $\ln 2 / 0.43 \approx 1.6$ years.

Typical values from the energy economics literature:

Sector	σ^{SR}	σ^{LR}	λ^{int}	$ \eta^{\text{SR}} / \eta^{\text{LR}} $	λ^{ext}
Road transport	0.2–0.5	1.0–2.5	0.10–0.25	0.1–0.3 / 0.3–0.7	0.15–0.45
Electricity gen.	0.1–0.3	0.5–1.5	0.05–0.25	0.05–0.1 / 0.1–0.3	0.15–0.50
Industry	0.1–0.4	0.5–1.5	0.10–0.30	0.1–0.2 / 0.3–0.5	0.20–0.40
Residential	0.1–0.3	0.3–1.0	0.10–0.30	0.05–0.2 / 0.2–0.5	0.15–0.40

6. Relationship to the TECP

The TECP (Paper I) is computed from the long-run equilibrium semi-elasticities, without partial adjustment. This is appropriate because:

1. The TECP is a *comparability metric*, not a forecast. It answers “what carbon tax is this policy equivalent to?” in the long run.
2. The adjustment speeds largely cancel in the TECP ratio, since both numerator and denominator are affected by the same frictions.
3. A time-dependent TECP would require choosing an arbitrary horizon, reducing comparability.

For year-by-year emissions projections, the partial adjustment model is applied in parallel: the CES computes equilibrium targets from policy-adjusted prices, and the partial adjustment filter determines actual year-by-year quantities and emissions. The difference between baseline and policy emissions paths gives the time profile of abatement, which converges to the TECP-implied long-run abatement as $t \rightarrow \infty$.

7. Implementation

7.1 CES form in a spreadsheet

The CES requires, per sector per year:

- One cell for C_t (the cost index, a SUMPRODUCT over fuels)
- One cell per fuel for $q_{f,t}^*$ (the equilibrium quantity)
- One cell per fuel for $q_{f,t}$ (the partially adjusted quantity)

For n fuels this is $2n + 1$ cells per sector per year, on top of the price cells already in the energy balance.

7.2 Isoelastic form in a spreadsheet

The isoelastic form replaces the cost index with a product over all fuel prices:

- One cell per fuel for $q_{f,t}^*$ (a PRODUCT formula over prices with elasticity exponents)
- One cell per fuel for $q_{f,t}$ (partial adjustment)

This is $2n$ cells per sector per year—one cell fewer than the CES (no cost index), but the individual formulas are longer because each fuel must reference every other fuel’s price.

7.3 Choice for CPAT-type applications

For tools like CPAT that already carry year-by-year energy balances with fuel-level prices and quantities, the CES form adds minimal overhead (one aggregation cell per sector per year) while providing guaranteed cross-elasticity consistency. The isoelastic form is preferable only when analytical simplicity matters more than dynamic accuracy—which is the case for the TECP formula itself, but not for the projection model.

Appendix: Complete Formula Reference

Throughout: fuel f in sector s , year t . Baseline (year 0) quantities q_{fs}^0 , prices p_{fs}^0 , emission factors e_f . CES substitution elasticity σ_s , demand elasticity η_s .

A.1 CES cost index

$$C_{s,t} = \left(\sum_f \alpha_{fs} p_{fs,t}^{1-\sigma_s} \right)^{1/(1-\sigma_s)} \quad (21)$$

where α_{fs} are calibrated share parameters from the base year.

A.2 CES equilibrium quantities

$$q_{fs,t}^* = q_{fs,t-1} \left(\frac{p_{fs,t}}{p_{fs,t-1}} \right)^{-\sigma_s} \left(\frac{C_{s,t}}{C_{s,t-1}} \right)^{\sigma_s + \eta_s} \quad (22)$$

A.3 Isoelastic equilibrium quantities

$$q_{fs,t}^* = q_{fs,t-1} \prod_g \left(\frac{p_{gs,t}}{p_{gs,t-1}} \right)^{\varepsilon_{fg,s}} \quad (23)$$

A.4 Partial adjustment (single speed)

$$q_{fs,t} = q_{fs,t-1} + \lambda_s (q_{fs,t}^* - q_{fs,t-1}) \quad (24)$$

A.5 Partial adjustment (split speeds)

$$Q_{s,t} = Q_{s,t-1} + \lambda_s^{\text{ext}} (Q_{s,t}^* - Q_{s,t-1}) \quad (25)$$

$$w_{fs,t} = w_{fs,t-1} + \lambda_s^{\text{int}} (w_{fs,t}^* - w_{fs,t-1}) \quad (26)$$

$$q_{fs,t} = w_{fs,t} \cdot Q_{s,t} \quad (27)$$

A.6 Elasticity matrix from CES

$$\varepsilon_{ff,s} = -\sigma_s + (\sigma_s + \eta_s) s_{fs} \quad (28)$$

$$\varepsilon_{fg,s} = (\sigma_s + \eta_s) s_{gs} \quad (g \neq f) \quad (29)$$

A.7 Semi-elasticities from CES

$$\tilde{\varepsilon}_{fs}^{\text{int}} = \sigma_s (1 - w_{fs}) \frac{e_f - \bar{e}_s}{p_{fs}} \quad (30)$$

$$\tilde{\varepsilon}_s^{\text{ext}} = |\eta_s| \frac{\bar{e}_s}{C_s} \quad (31)$$

A.8 Adjustment speed from empirical elasticities

$$\lambda \approx \sigma^{\text{SR}} / \sigma^{\text{LR}} \quad (32)$$

$$\sigma = \sigma^{\text{LR}} \quad (33)$$

$$\sigma_T^{\text{eff}} = \sigma [1 - (1 - \lambda)^T] \quad (34)$$

$$t_{1/2} = \ln 2 / \ln(1/(1 - \lambda)) \approx \ln 2 / \lambda \quad (35)$$